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House Price Prediction Using Linear Regression and Housing Dataset

1. INTRODUCTION

House price prediction is another practical application of Machine Learning that aids in calculating the actual worth of a property based on its features. Prediction of the right amount will be useful for home buyers, sellers, real estate organizations, and investors alike.

The objective of this project is to build a model based on the linear regression algorithm using Python to predict the prices of houses from some important features including the area, number of bedrooms, bathrooms, stories, and parking spaces. This project will be carried out using a dataset of the housing data consisting of real estate data.

The following are the Python packages used in this project: Pandas, Matplotlib, and Scikit-learn.

2. PROBLEM STATEMENT

The objective of this project is to build a Machine Learning model that can predict the price of a house based on selected features.

The project aims to:

- Analyze the housing dataset.
- Identify important factors affecting house prices.
- Train a Machine Learning model using Linear Regression.
- Predict the price of new houses using the trained model.

- Evaluate the relationship between house features and prices.

3. METHODOLOGY

Step 1: Import Required Libraries

Libraries used:

- Pandas
- Matplotlib
- Scikit-learn

Step 2: Loading of Dataset

The Housing.csv dataset was loaded into a Pandas DataFrame.

Step 3: Data Exploration

- Displayed first five records.
- Checked for missing values.
- Selected important columns.

Step 4: Data Visualization

A scatter plot was created between:

- House area
- House price

This helped visualize the positive relationship between area and price.

Step 5: Model Building

Linear Regression was used.

The dataset was divided into:

- Training data (80%)
- Testing data (20%)

The model was trained using the selected features.

Step 6: Prediction

The model predicted prices for:

- Test data
- A new house with given specifications.

5. TOOLS AND TECHNOLOGY USED

Tool	Purpose
Python	Programming language
Pandas	Data manipulation
Matplotlib	Data visualization
Scikit-learn	Machine Learning model
Jupyter Notebook	Development environment

6. RESULTS AND DISCUSSION

Observations

1. House area has a positive relationship with price.
2. Houses with more bedrooms and bathrooms generally have higher prices.
3. Parking availability also influences house prices.
4. Linear Regression successfully identified relationships between input features and house prices.

Advantages

- Easy to implement.
- Fast prediction.
- Useful for preliminary real estate price estimation.

Limitations

- Only a few features were used.

- Other important factors such as location, furnishing status, and air conditioning were ignored.
- Linear Regression assumes a linear relationship, which may not always be accurate.

7. CONCLUSION

In conclusion, the House Price Prediction System utilizing Linear Regression from Python language has been executed effectively. The algorithm was developed using critical characteristics of houses and could accurately predict house prices given specific details. Machine Learning makes it possible to make simple real-world issues such as predicting house prices easy to solve. In addition, accurate prediction in the future can be achieved by incorporating more variables into the algorithm and using complex Machine Learning models like Random Forest, Decision Tree etc.